

Intro

Recsys on large e-commerce platforms have a fundamental tension area → users who need personalised recommendations are the one with least data.

- From our Amazon findings, 72% of Beauty & Personal Care shoppers have < 5 reviews in that category.
- This makes within category collab filtering nearly useless for majority of user base

However, these users are not truly unknown, as their behaviour can be captured in other categories.

- Clothing purchases: aesthetic taste
- Sports & Outdoors: lifestyle orientation
- Cross category signals exist, but just underutilised

We aim to build a multi signal recommender system that pushes relevant Beauty product recommendations to users **earlier** → before they have accumulated enough within category history for a traditional model to function.

To do so, we aim to combine 4 complementary info sources:

1. Cross category collaborative filtering: user item interaction across beauty, clothing, sports
2. Visual style embeddings: CV embeddings from Clothing product images
3. Sequential purchase patterns: temporal cross category sequences
4. Semantic item features: enrich product descriptions with OCR/LLM based enrichments

Each component will address a different aspect of the cold start problem, and together, this forms a richer user representation (rather than just any 1 signal)

Dataset

- Dataset:
 - [McAuley-Lab/Amazon-Reviews-2023 · Datasets at Hugging Face](#)
 - 571.54M user-product reviews on Amazon collected in 2023 by [McAuley Lab](#)
 - Reviews are grouped by *category*

Category	Purpose	Dataset Size
Beauty	Target domain – what we recommend	~ XX reviews
Clothing	Source domain 1 → style / aesthetic signal	~XX reviews
Sports & outdoors	Source domain 2 → lifestyle	~XX reviews

	/ wellness signal	

Each review record contains:

Each item metadata record contains:

Methodology of Collection

Wut is this????

We access data by direct JSONL streaming from HF Hub using Python requests library???

We serve files from:

/resolve/main/raw/review_categories/{Category}.jsonl: reviews

/resolve/main/raw/meta_categories/meta_{Category}.jsonl: metadata

We implemented a custom streaming loader that reads line by line and stops at a configurable sample ceiling, to make the pipeline reproducible without downloading the full dataset (750GB.....). Files are publicly hosted

Expected Data Size

Subset	Rows	Approx Size
Reviews (3 categories * XXX k)		
Item metadata (3 * XXX k)		
Product images (sampled for the CV stuff)		

For full scale training in later experiments, the complete beauty category can be streamed later on???

Data Augmentation

We don't augment review interaction data

- But we do enrich item representations

- CV pipeline: download clothing product images through the metadata links, extract visual embeddings (how???? Pretrained CLIP? etc)
- OCR / LLM pipeline: parse features + description fields? Apply XX-based structured extraction to produce normalised item attribute tags (e.g. material / skin type / use case) for beauty stuffs
- These enrichments are additional → base CF model trains on the raw interaction data, and then we inject these enriched features as item-side info?
- **I think idk**

Recommendation Problem

Formal definition

Given:

- User u with interaction history in clothing and/or sports
- Target category in beauty, where user u has few/no interactions

Produce: ranked list of beauty items likely to be relevant to user u

GABRIEL DO YOUR LATEX THINGY?!

Cold start framing

We define cold users as those with < 5 beauty interactions in training set

- EDA confirms this isn't edge case:
- Out of 93,012 users in Beauty from 500k samples,
 - Cold: 66,964 (72%)
 - Warm in clothing/sports: 15,369 (~23% of cold)
 - Cold everywhere: 51,595 (~77% of cold)

For the 23% of cold users with the cross domain behaviour signals, our system can use the CF transfer / sequential models. For the remaining 77%, item side features (CV, OCR etc) can provide a content based fallback?

Why problem is significant

1. Scale of problem:
 - With 72% of beauty users being cold, within category CF system is poopoo for majority of audience. On platform like amazon's scale, prolly millions of missed personalisation opps every day
2. 246 day window:

- a. Median / mean days... large observable window where we could be pushing beauty recommendations. But rn cannot with a silo model, as there is no beauty history
- 3. Category entry patterns
 - a. For users with all 3 categories, 58% entered through clothing first (10.8k out of 22k) → not beauty
 - b. Most natural onboarding for beauty shopper is through clothing
 - c. Clothing: highest value source domain for transfer
- 4. Cross domain correlations are real but (i think) not trivial
- 5. Generalisability
 - a. Platform agnostic → every multi category retailer likely faces this cold start structure
 - b. Working cross category cold start system will have direct commercial value

Evaluation Framework (overview)

Evaluation Philosophy / Principles

- Central claim: cross category signals improve recommendation quality for cold Beauty(?) users
- So every design choice will be to test that hypo

Some design principles:

1. Cold users are the primary target:
 - a. 72% of Beauty users have <5 reviews
 - b. Aggregate metrics would otherwise be dominated by the 28% of warm users (which any CF model can prolly predict reasonably well)
 - c. Business value: these cold users are the ones who would churn under normal models? New shoppers who receive irrelevant Beauty recommendations on first visit is unlikely to return → our cross domain transfer will address this specifically.
 - d. **246 day window?** (need to re see potentially): EDA shows that these category users make the first cross-category purchase a median of 246 days before their first Beauty purchase
 - e. This large window of observable behaviour (aka when other Clothing etc purchases are accumulating) would be where a cold start aware model could already be serving personalised recommendations
2. No data leakage
 - a. Temporal split → train on purchases before a cutoff, test on purchases after
 - b. Ensure you don't see the future
3. Measure the incremental contribution of each component:
 - a. We add each component 1 at a time (on top of the previous stage) → CF, CV embeddings, sequential, semantic - or some order of it.
 - b. This allows us to also evaluate marginal value of each signal explicitly (compared to just evaluating the final model)

Data Splits

Temporal Train / Test split

- All review interactions ordered by timestamp
- TBC: we apply single global cutoff **based on XXX** of timestamps in the data
- Train: all interactions before cutoff timestamp of XXX → model fitting, embedding learning, graph construction etc etc
- Test: held out interactions after cutoff timestamp → 1 +ve per test user = their most recent? Beauty??? Purchase
 - I am thinking, for each user → we hold out 1 (or N) of their beauty purchases (aka the most recent), which is what the model must find.
 - The model should then rank all the candidate items, and check where that 1 held out item appears in the ranking????
 - Why 1: temporal split likely to give only 1 (most recent beauty purchase after cutoff)
 - Limitation: with just 1 positive per user, recall @ K will become binary (either found it or didn't, 1/0). This might make evaluation noisy for individual users, but will likely average out for all users????
- This temporal split ensures models are not trained on a user's future purchases (which creates leakage)
 - The model would otherwise learn what users **will** buy, rather than what they **would want to buy** (given only their history)
 - So this temporal split enforces that the model can only see info it already has available at the time of recommendations

Sequential models

[Claude cuz I dont really understand oops]

For SASRec based sequential component, Claude recommends to additionally apply a leave-one-out split, scoped to each user's Beauty interaction subsequence

interaction subsequence:

```
'''
Test positive:  last Beauty item in the user's cross-category sequence
Training signal: all prior events (cross-category), including earlier
Beauty interactions
'''
```

```
This is the standard evaluation protocol for sequential recommendation
(used in SASRec, BERT4Rec). It ensures the sequential model is tested
on predicting the next Beauty item from the user's temporal context –
the exact inference-time task.
```

Train/Val/Test

For hyperparameter tuning during training, hold out a val set from training period:

- Training: reviews before (cutoff - 90 days)???
- Val: reviews in (cutoff - 90 days, cutoff)
- Test: reviews after cutoff

We report final metrics on test set only.

User partitions

- Cold users: 72% of users → primary eval
 - Users with beauty(?) count in train set of <5
- Warm users: 28%
 - Users with beauty(?) count in train set of >= 5

Why 5: might need to adjust

Beauty train set interactions	% of users	CF quality
0	49	No CF signal at all
1-4	23	Extremely sparse → CF unreliable?
5-9	8	CF likely able to begin functioning
10+	20	Sufficient for within category CF?

Users with 0-4 Beauty interactions cannot be served well by any within-category model.

- They are the direct target of cross-domain transfer.
- 5-interaction threshold captures this group?

How to report:

Partition	Users	Role
All	100%	Overall system performance
Cold	~72%?	Primary: tests cross domain transfer claims
Warm	28%	Secondary: to ensure that there's no regression for existing users?

Model is considered to advance / help if it improves the cold user metrics (without regressing too much for warm?) → warm user improvement is nice to have, but secondary?

Metrics:

1. NDCG@K:
 - a. Rewards model for placing relevant item higher in the list
 - b. Rank 3rd better than like ranking 10th → we should catch this distinction
 - c. Right metric when users look at results sequentially
 - d. Why @10/@50:
 - i. @10: typically 1 screen of results have ~10 recommendations (like a typical recommendation widget)
 - ii. @50: captures broader recall quality → whether relevant item appears anywhere in an extended list (aka user will catch if he just scrolls etc)
 - iii.
2. Recall@K:
 - a. With a single test positive per user, Recall@K is a binary hit/miss indicator: did the relevant item appear in the top K?
 - b. When we average across users, it gives the fraction of users for whom the model surfaces the correct item within the top K
 - c. Why recall in addition to NDCG:
 - i. NDCG measures ranking quality, while recall measures coverage.
 - ii. A model with high NDCG but low Recall is precise but narrow → it will rank the relevant item high when it appears, but often misses it entirely.
 - iii. For cold users with very sparse signals, we may accept lower position precision in exchange for not missing the item altogether?

Metric	What it means
NDCG@10	Is the relevant item near the top of a short list?
Recall@10	Does the model surface the item in the top 10 at all?
NDCG@50	Is ranking quality preserved over a longer list?
Recall@50	For items the model struggles to rank highly, does it at least include them in a wider set?

Negative Sampling

For top-k eval, we score the test positive against a pool of negative items. Depending on model type / computational feasibility, we can look at:

1. **Full ranking**: score test item against all Beauty items that are not seen in training. The model must surface the test item from the full item pool
2. **Sampled negatives** (for seq models / if too intensive): score the test item against like 99 randomly sampled unobserved items → so you can a 100-item evaluation (can prob back up with a paper)

AI Overview

In sequential Recommender Systems (RecSys), the **100-item evaluation protocol** (often called "leave-one-out" or "next-item ranking") is the standard benchmark method. It evaluates how accurately a model can rank the user's next true interaction against 99 randomly sampled negative items.  ACM Digital Library +1

Core Protocol Mechanics

1. **The Target**: For each user, the *most recent* historical interaction is held out as the ground-truth positive item.
2. **The 100 Items**: The model is presented with this true item, mixed with **99 randomly sampled candidate items** (items the user has *never* interacted with in their sequence).
3. **The Ranking**: The model scores all 100 items by relevance and ranks them.  ACM Digital Library +2

Primary Evaluation Metrics

Once the model ranks the 100 items, success is typically measured using these ranking metrics:

- **Hit Rate @ K (HR@K)**: Checks if the true item is in the top K positions of the ranked list (e.g., HR@5 or HR@10).

$$\text{HR@K} = \frac{\text{Number of Hits in Top } K}{\text{Total Number of Items}} \quad \text{ACM Digital Library +1}$$

- Report like the NDCG@10/Recall@10 within sampled pool

If both run, then we report both figures (sampled just to support?)

Cold Start Recovery Rate

- Apart from standard ranking metrics, can report **cold start recovery rate** → fraction of cold users that the model(s) can surface a relevant item within like top 50 (vs how many WMF or sth would have found)
- This helps quantify value of this model / improvements → business interpretable number
 - X% of users who would not have received any relevant recommendations will now receive at least 1

Baseline Hierarchy

We arrange baselines in a hierarchy that will mirror a N-stage system archi → each stage adds 1 type of signal

- This is to answer the key qn of **what does each signal add?**

AI cooked this:

```
Stage 0: WMF (Beauty only)
    ↓ add cross-category CF
Stage 1: LightGCN_B+C+S (unified graph)
    ↓ add visual style embeddings
Stage 2: LightGCN_B+C+S + CLIP item features
    ↓ add temporal sequence modelling
Stage 3: CF candidates + SASRec re-ranker
    ↓ add semantic item content
Stage 4: Full system (CF + CV + Seq + LLM features)
```

OTHER STUFF AI COOKED LAST NIGHT FOR PREAMBLE, MIGHT BE RELEVANT FOR EXPERIMENTS? IDK

Within-Category Baselines ("Silo Models")

These represent what any standard production recommender does – train and serve recommendations entirely within one category.

Model	Description	Rationale
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WMF	Weighted Matrix Factorisation on Beauty only	Standard implicit CF; the primary baseline to beat
EASE	Closed-form linear autoencoder on Beauty only	Strong shallow baseline; hard to beat with small data
LightGCN_B	Graph CF on Beauty-only user-item graph	State-of-art CF baseline; best within-category model

Why WMF specifically as the primary baseline? WMF is the closest analogue to what many production systems run. It is robust, interpretable, and well-understood. It gives a fair upper bound on what a category-silo system can achieve, making the cross-category gain maximally credible.

5.2 Stage 1 – Cross-Category CF Ablations

Model	Categories used	Hypothesis
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LightGCN_B+C	Beauty + Clothing	Does Clothing transfer help?
LightGCN_B+S	Beauty + Sports	Does Sports transfer help?
LightGCN_B+C+S	All three	Do they complement each other?

EMCDR (C→B) Clothing → Beauty mapping Explicit embedding transfer; precision focus
EMCDR (S→B) Sports → Beauty mapping Explicit embedding transfer; coverage focus
EMCDR (C+S→B) Combined mapping Best of both source domains?

5.3 Stage 2 – + Visual Features

Model Description
--- ---
LightGCN_B+C+S + CLIP Unified graph with CLIP image embeddings as item feature initialisation

5.4 Stage 3 – + Sequential Re-ranking

Model Description
--- ---
SeqBase SASRec on Beauty interactions only
SeqCross SASRec on cross-category sequences + category token
SeqCross_LT SeqCross + long-term user embedding ($\alpha \cdot h_t + (1-\alpha) \cdot u_{lt}$)
SeqCross_recent SeqCross, last 6 months only (truncation ablation)
CF+Seq LightGCN_B+C+S candidates → SeqCross_LT re-ranker

5.5 Stage 4 – + Semantic Features

Model Description
--- ---
Full system Best CF model + CLIP + SeqCross_LT + LLM-extracted item attributes

6. Experiment Matrix

Each row in this table is a separately run and evaluated experiment. The "Claim tested" column links back to a specific hypothesis; models that do not advance any claim are not run.

ID Model Partition Claim Tested
--- --- --- ---

| E0 | WMF_B, EASE_B, LightGCN_B | All / Cold / Warm | Upper bound of silo CF |

| E1 | LightGCN_B+C, B+S, B+C+S | All / Cold / Warm | H1: Cross-category CF > silo on cold users |

| E2 | EMCDR(C→B), (S→B), (C+S→B) | All / Cold / Warm | H2: Clothing = precision, Sports = coverage |

| E3 | LightGCN_B+C+S + CLIP | All / Cold / Warm | H3: Visual features improve item-side matching |

| E4 | SeqBase, SeqCross | Cold / Warm | H4-Seq-1: Cross-domain sequences > within-domain |

| E5 | SeqCross_LT vs SeqCross | Cold / Warm | H4-Seq-2: Long-term component fixes recency bias |

| E6 | SeqCross_recent vs SeqCross | Cold / Warm | H4-Seq-3: Recency truncation hurts |

| E7 | CF+Seq vs LightGCN alone | Cold / Warm | H4-Seq-4: Sequential re-ranking adds temporal precision |

| E8 | Full system | All / Cold / Warm | End-to-end system vs best single-signal model |